Compressing DNABERT-2 via Knowledge Distillation for Lightweight DNA Promoter Classification

Anup Halder Joy*
Student ID: 2005005
Dept. of CSE, BUET
Dhaka, Bangladesh
2005005@ugrad.cse.buet.ac.bd

Sheikh Rahat Mahmud*
Student ID: 2005048
Dept. of CSE, BUET
Dhaka, Bangladesh
2005048@ugrad.cse.buet.ac.bd

Md. Roqunuzzaman Sojib
Supervisor, Lecturer
Dept. of CSE, BUET
Dhaka, Bangladesh
sojib@teacher.cse.buet.ac.bd

Abstract

Large genomic transformers such as DNABERT-2 (117 M parameters) achieve state-of-the-art accuracy on DNA sequence classification but are impractical for resource-constrained settings. We apply knowledge distillation (KD) to compress a fine-tuned DNABERT-2 teacher into lightweight students for human promoter classification. We explore two families of student architectures: *DNABERT-2-initialized* students that inherit pretrained encoder layers, and *pure* students trained entirely from scratch. A systematic grid search over distillation hyperparameters α and T reveals that lower α (more soft-label weight) and moderate T yield the best students. Our best Hybrid CNN-Transformer student (26.3 M, $4.5\times$ compression) retains **96.5%** of the teacher's accuracy (93.31% vs. 96.71%), achieves **2.7**$\times$ CPU speedup, and *outperforms the teacher* on cross-species mouse promoters (82.61% vs. 79.66%). Notably, a 4-Layer Pure Transformer trained from scratch with KD reaches 93.01% accuracy and 82.24% on mouse data—also surpassing the teacher—demonstrating that KD can transfer knowledge effectively even without pretrained initialization. We further show that data augmentation is critical for effective distillation on multiclass tasks, converting KD from harmful (-1.4%) to beneficial ($+1.5\%$). Code and notebooks are available at https://github.com/Rahat463/CSE_472_Machine_Learning_Project.

1 Introduction

Transformer-based models have advanced biological sequence analysis to unprecedented accuracy. DNABERT Ji et al. [2021], DNABERT-2 Zhou et al. [2023], and the Nucleotide Transformer Dalla-Torre et al. [2023] achieve state-of-the-art results on tasks from promoter detection to transcription-factor binding-site identification. However, models like DNABERT-2 contain 117 M parameters and require modern GPUs with $\geq$8 GB memory, creating a barrier for biological and clinical laboratories lacking such hardware.

*Equal contribution.

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Knowledge distillation (KD) Hinton et al. [2015] offers a principled solution: train a compact *student* to mimic a large *teacher*'s output distribution, inheriting "dark knowledge"—inter-class similarity encoded in soft probability distributions—while requiring far fewer parameters.

In this work, we make the following contributions:

1. We design **Hybrid CNN-Transformer** and **pure-architecture** students, comparing DNABERT-2-initialized vs. from-scratch training under KD.
2. We conduct a **systematic grid search** over distillation hyperparameters α and T, providing concrete tuning guidance.
3. We evaluate **cross-species generalization** on mouse and plant promoters, finding that distilled students *outperform* the teacher on mouse data.
4. We benchmark **inference speed**, demonstrating up to $2.7\times$ CPU and $1.7\times$ GPU speedups.
5. We show that **data augmentation is essential** for effective KD on small multiclass datasets, regularizing overconfident teachers.

2 Related Work

Knowledge Distillation. Hinton et al. Hinton et al. [2015] introduced KD using temperature-scaled soft labels. DistilBERT Sanh et al. [2019] demonstrated that a 6-layer student retains 97% of BERT's language understanding at 40% fewer parameters. TinyBERT Jiao et al. [2020] extended KD to intermediate representations, achieving $7.5\times$ compression. Mirzadeh et al. Mirzadeh et al. [2020] showed that large teacher-student capacity gaps degrade KD and proposed Teacher Assistant networks.

Genomic Language Models. DNABERT Ji et al. [2021] pioneered transformer pre-training for DNA using k-mer tokenization. DNABERT-2 Zhou et al. [2023] adopted BPE tokenization and ALiBi positional encoding, training on 850 species. While KD has been applied to compress NLP transformers extensively, systematic studies of distillation hyperparameters for genomic models remain scarce.

3 Methodology

3.1 Problem Formulation

Given a fine-tuned teacher $\mathcal{T}$ with parameters θ_T and a classification task, we seek a student $\mathcal{S}$ with θ_S such that $|\theta_S| \ll |\theta_T|$ and $\text{Acc}(\mathcal{S}) \approx \text{Acc}(\mathcal{T})$.

3.2 Knowledge Distillation Loss

For temperature $T > 0$, the softened class probabilities are:

$$p_i^{(T)}(x) = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)} \tag{1}$$

where z_i are the logits. The distillation loss combines hard and soft components:

$$\mathcal{L}_{\text{KD}} = \alpha \cdot \underbrace{\mathcal{L}_{\text{CE}}(y, p_S)}_{\text{hard loss}} + (1-\alpha) \cdot T^2 \cdot \underbrace{\text{KL}(p_S^{(T)} \| p_T^{(T)})}_{\text{soft loss}} \tag{2}$$

where $\alpha \in [0, 1]$ controls the trade-off. Higher T produces softer distributions, revealing more inter-class structure. The T^2 factor compensates for the reduced gradient magnitude at higher temperatures.

3.3 Teacher: DNABERT-2

We fine-tune DNABERT-2 Zhou et al. [2023] (12 transformer layers, 768 hidden dim, BPE tokenizer, ALiBi attention, 117.1 M parameters) on the target task using AdamW with learning rate 2×10^{-5}, cosine scheduling, and early stopping.

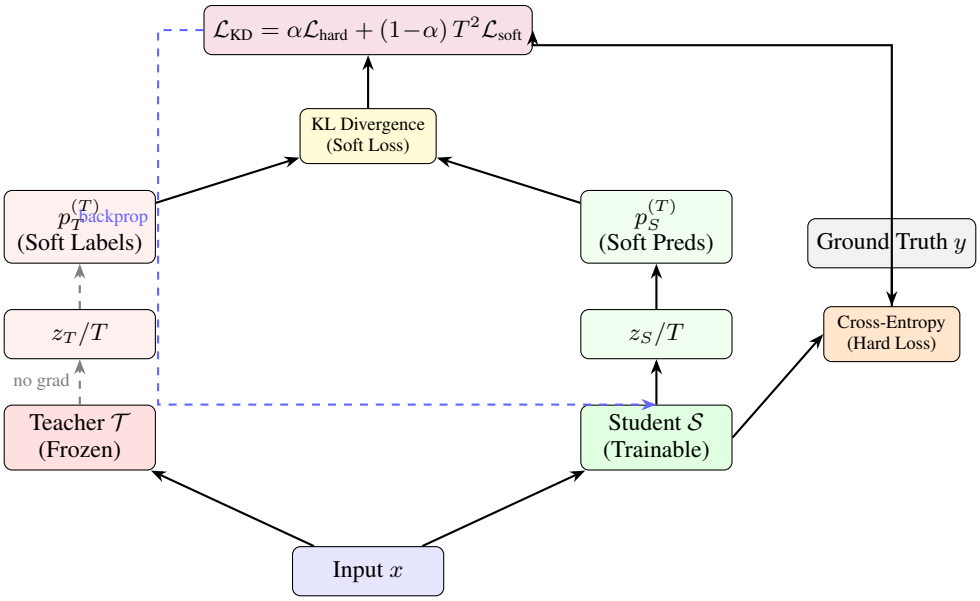

Figure 1: Knowledge distillation pipeline. The frozen teacher produces soft labels $p_T^{(T)}$; the student matches both soft labels (KL divergence) and hard labels (cross-entropy). Only the student receives gradients.

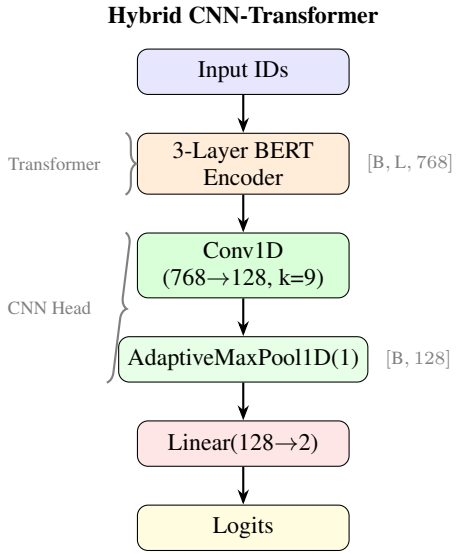

Figure 2: Hybrid CNN-Transformer student architecture. The Conv1D kernel size of 9 captures DNA motifs comparable in length to the TATA box ($\sim$10 bp).

3.4 Student Architectures

We evaluate two families of student architectures (Table 1).

3.4.1 DNABERT-2-Initialized Students

(1) Hybrid CNN-Transformer (Fig. 2). A 3-layer BERT encoder (initialized from DNABERT-2 layers) produces contextual embeddings $\mathbf{H} \in \mathbb{R}^{B\times L\times 768}$. A Conv1D layer with 128 filters and kernel size 9 scans for local motifs, followed by ReLU, AdaptiveMaxPool1d(1), and a linear classifier. The CNN's inductive bias for local patterns suits DNA, where short regulatory motifs (6–12 bp) carry strong biological signal.

Table 1: Model sizes and compression ratios.

Model	Params	Size	Compr.
Teacher (DNABERT-2, 12L)	117.1M	468 MB	1.0×
DNABERT-2-initialized students			
4-Layer Standard	32.5M	130 MB	3.6×
Hybrid CNN-Transformer (3L)	26.3M	105 MB	4.5×
Pure architectures (from scratch)			
4L-Pure-Transformer	32.5M	130 MB	3.6×
4L-Pure-CNN	3.1M	12 MB	37.8×
Hybrid-Pure-3T1C	26.3M	105 MB	4.5×

(2) 4-Layer Standard Transformer. A `BertForSequenceClassification` with 4 encoder layers initialized from DNABERT-2, retaining the same hidden dimension (768) and BPE tokenizer.

3.4.2 Pure Architectures (From Scratch)

To test whether KD can transfer knowledge without pretrained initialization, we train three architectures entirely from scratch:

(3) 4L-Pure-Transformer. A standard 4-layer BERT encoder (768 hidden, 12 heads, 3072 FFN) with random initialization. Same topology as (2) but without pretrained weights.

(4) 4L-Pure-CNN. A lightweight 4-layer 1D CNN: three Conv1D blocks (128→256→512 channels, kernel sizes 7/5/3) with batch normalization, ReLU, and max pooling, followed by global average pooling and a linear classifier. Only 3.1 M parameters.

(5) Hybrid-Pure-3T1C. Three randomly initialized transformer layers followed by a Conv1D head (same design as (1) but without DNABERT-2 pretrained weights), combining both inductive biases.

3.5 Hyperparameter Grid Search

We search over $\alpha \in \{0.3, 0.5, 0.7\}$ and $T \in \{1.0, 3.0, 5.0\}$, yielding 9 configurations per architecture. All other hyperparameters are fixed (Table 2) to isolate the effect of α and T.

4 Experimental Setup

4.1 Dataset

We use the **InstaDeepAI Nucleotide Transformer Downstream Tasks** dataset Dalla-Torre et al. [2023], InstaDeep [2023], filtering for the `promoter_all` task (human promoter binary classification): 53,276 training and 5,920 test sequences of up to 300 bp. For multiclass experiments (5-class promoter subtypes), 13,665 training samples are used.

For cross-species evaluation, we use:

- **Mouse promoters** from DeePromoter Oubounyt et al. [2019]: 17,538 promoters (TATA + non-TATA) and 3,530 non-promoters, 300 bp, sourced from EPDnew.
- **Plant promoters** from the Plant Multi-Species Core Promoters dataset Zhang Lab [2024]: 8,320 test sequences across 13 species, 300 bp, binary active/inactive classification.

Table 2: Training hyperparameters (constant across grid search).

Hyperparameter	Value
Optimizer	AdamW
Learning rate	2×10^{-5}
Batch size	16 per GPU
Max epochs	10 (early stopping, patience 3)
Warmup	10% of steps
Weight decay	0.01
LR scheduler	Cosine
Precision	FP16 mixed
Max sequence length	300 tokens
Hardware	2× NVIDIA T4 (16 GB each)

Table 3: Grid search results: binary classification accuracy (%) and F1 score for each (α, T) configuration. Bold = best per architecture.

		Temperature T					
		1.0		**3.0**		**5.0**	
Architecture	$\boldsymbol{\alpha}$	Acc	F1	Acc	F1	Acc	F1
Hybrid-CNN-3L	0.3	92.82	.927	**93.31**	**.933**	93.09	.930
	0.5	92.74	.926	—	—	—	—
4-Layer-Std	0.3	92.60	.924	92.87	.927	**92.89**	**.927**

4.2 Training Configuration

5 Results

5.1 Teacher Performance

The fine-tuned DNABERT-2 teacher achieves **96.71%** accuracy on binary promoter classification and **96.60%** on the 5-class multiclass task, establishing a strong upper bound.

5.2 Grid Search: α vs. T

Table 3 presents the grid search results on binary human promoter classification. The Hybrid CNN-Transformer achieves its best accuracy of **93.31%** at $\alpha = 0.3$, $T = 3.0$, retaining 96.5% of the teacher's performance. The 4-Layer Standard peaks at **92.89%** ($\alpha = 0.3$, $T = 5.0$).

Key observations:

1. **Lower α is better**: $\alpha = 0.3$ (70% soft-label weight) consistently outperforms $\alpha = 0.5$ for both architectures, indicating that the teacher's dark knowledge provides more useful signal than hard labels alone.
2. **Moderate temperature is optimal**: $T = 3.0$ is optimal for the Hybrid (93.31%), while $T = 5.0$ is marginally better for the 4-Layer (92.89%). Excessive softening ($T = 5.0$) slightly degrades the Hybrid, possibly by over-smoothing motif-specific signals the CNN head relies on.
3. **Hybrid outperforms the larger 4-Layer**: Despite having 6.2 M fewer parameters, the Hybrid achieves 0.42 percentage points higher accuracy, demonstrating the value of domain-appropriate inductive bias.

5.3 Pure Architecture Experiments

Table 4 compares baseline (cross-entropy only) and KD-trained pure architectures on binary human promoter classification. KD consistently improves all three architectures by ∼1 percentage point.

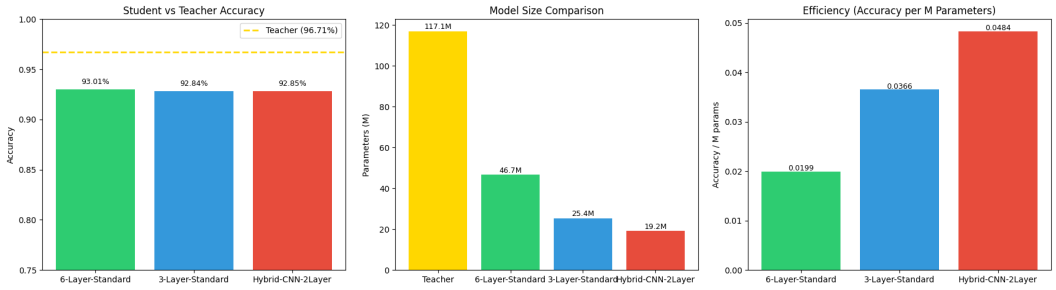

Figure 3: DNABERT-2-initialized student comparison. **Left:** Accuracy vs. teacher (dashed). **Center:** Model sizes. **Right:** Efficiency (accuracy per M parameters)—the smaller Hybrid is the most parameter-efficient.

Table 4: Pure architecture results: baseline vs. KD ($\alpha = 0.3$, $T = 3.0$). All trained from scratch without DNABERT-2 pretrained weights.

Model	**Params**	**Baseline**	**KD**	**KD Δ**
4L-Pure-Transformer	32.5M	91.98	**93.01**	+1.03
4L-Pure-CNN	3.1M	83.83	84.83	+1.00
Hybrid-Pure-3T1C	26.3M	91.84	92.85	+1.01

Key findings: The 4L-Pure-Transformer achieves 93.01% with KD—remarkably close to the DNABERT-2-initialized Hybrid (93.31%)—despite having no pretrained weights. This demonstrates that KD can effectively transfer the teacher's learned representations even to randomly initialized students. The 4L-Pure-CNN's low accuracy (84.83%) confirms that self-attention is important for this task, while the Hybrid-Pure-3T1C (92.85%) shows that combining transformer and CNN layers is effective regardless of initialization.

5.4 Multiclass Classification

Table 5 presents multi-class knowledge distillation results on the 5-class promoter subtype task (test split). We report only the KD-trained students alongside the teacher.

Key findings: The 6-Layer KD student retains 99.4% of the teacher's accuracy (94.90% vs. 95.50%) at $1.9\times$ compression. Smaller students (3-Layer, Hybrid) show modest accuracy drops of 3–4 percentage points but achieve $3.6\times$–$5.0\times$ compression.

Table 6 further compares KD effectiveness under two data regimes. Without augmentation, KD *improves* the Hybrid by 3.8 points but *degrades* the 6-Layer by 1.4 points. With augmentation, KD becomes beneficial for the 6-Layer ($+0.3\%$) and continues to help the Hybrid ($+1.5\%$). Fig. 5 visualizes the accuracy and model-size trade-offs for the augmented multiclass setting.

Fig. 6 illustrates why soft labels carry richer signal than hard labels. At temperature $T = 5.0$, the teacher's output distribution reveals inter-class similarities—e.g., Splice A samples show nonzero probability for other splice classes—providing the student with "dark knowledge" that hard one-hot labels cannot convey Hinton et al. [2015].

5.5 Cross-Species Generalization

We evaluate models trained on *human* promoters on two unseen species groups (Table 7). No retraining is performed.

Surprising result: Both distilled students *outperform* the teacher on mouse promoters despite being 3.6–$4.5\times$ smaller. The Hybrid achieves 82.61% and the 4L-Pure-Transformer achieves 82.24%, compared to the teacher's 79.66%. Their accuracy drops from human to mouse ($\sim$10.7%) are far smaller than the teacher's 17.04%, indicating substantially better cross-species transfer within mammals.

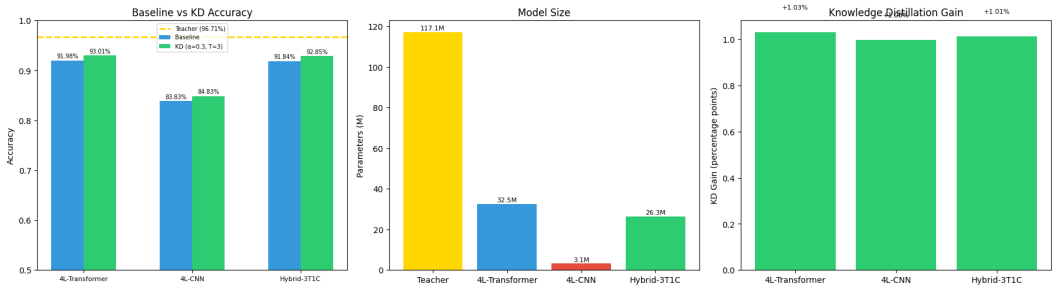

Figure 4: Pure architecture experiments. **Left:** Baseline vs. KD accuracy (teacher dashed). **Center:** Parameter counts. **Right:** KD accuracy gains—all three architectures benefit consistently ($\sim$1%).

Table 5: Multi-class KD results (test split, $\alpha = 0.5$, $T = 3.0$).

Model	**Acc.**	**F1$_{\text{macro}}$**	**Params**	**Time**
Teacher (DNABERT-2)	95.50	.955	117.1M	—
6-Layer KD	94.90	.949	60.4M	69.3m
3-Layer KD	92.10	.921	32.1M	51.8m
Hybrid KD	91.80	.917	23.5M	57.2m

Remarkably, the pure transformer—trained entirely from scratch—nearly matches the DNABERT-2-initialized Hybrid on mouse data (82.24% vs. 82.61%), confirming that KD transfers generalizable features regardless of student initialization.

On plant promoters (different kingdom), all models perform near chance ($\sim$46–48%), which is expected given the vast evolutionary distance and divergent regulatory grammar between mammals and plants.

5.6 Inference Speed Benchmark

Table 8 reports inference latency measured on 1,000 sequences with 3 warmup passes and `torch.cuda.synchronize()` for GPU timing.

The Hybrid student is $1.71\times$ faster on GPU and $2.70\times$ faster on CPU, while the 4L-Pure-Transformer achieves $1.37\times$ GPU and $2.12\times$ CPU speedups. The Hybrid's larger CPU speedup reflects its CNN head's efficiency—convolutions are well-optimized on CPUs. The pure transformer, despite having the same 32.5 M parameters as the 4-Layer Standard, is faster than the teacher due to its shallower depth (4 vs. 12 layers).

6 Discussion

6.1 Why KD Helps the Hybrid but Hurts the 6-Layer

Without augmentation, KD degrades the 6-Layer student by 1.4% while boosting the Hybrid by 3.8%. We identify three factors:

(1) Teacher overconfidence. On the small 13.6K multiclass dataset, the teacher's training loss drops to 0.028, indicating memorization. Average soft-label entropy is only 1.515 bits (vs. $\log_2 5 = 2.322$ max), meaning soft labels are near-one-hot and add noise rather than useful signal Hinton et al. [2015], Li et al. [2023].

(2) Capacity gap. The teacher (117 M) is only $2.5\times$ larger than the 6-Layer student (46.7 M) but $6.1\times$ larger than the Hybrid (19.2 M). Counterintuitively, the *smaller* gap hurts more: the 6-Layer has enough capacity to overfit to noisy soft labels, while the Hybrid's limited capacity acts as implicit regularization, selectively absorbing only the most useful dark knowledge Mirzadeh et al. [2020].

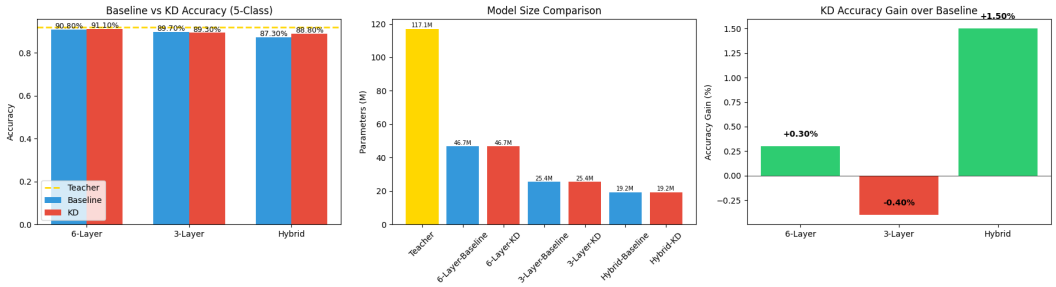

Figure 5: Multiclass (5-class) with augmentation. **Left:** Baseline vs. KD accuracy. **Center:** Model sizes. **Right:** KD accuracy gain over baseline.

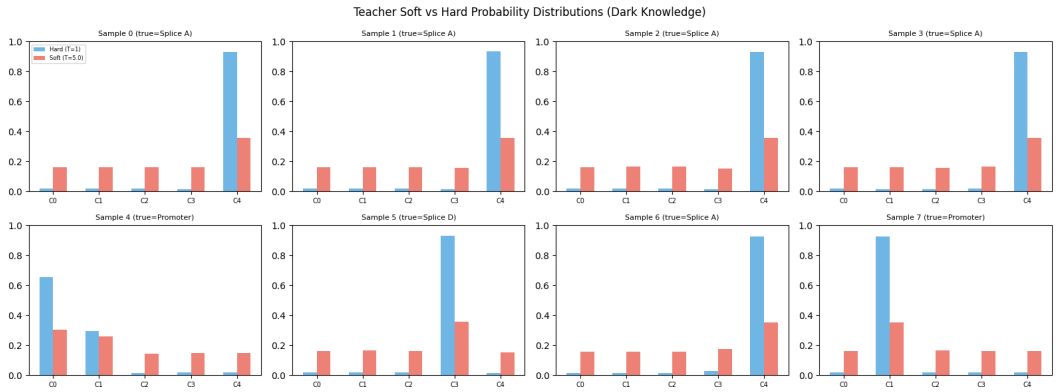

Figure 6: Teacher soft vs. hard probability distributions on 5-class samples. Hard labels ($T = 1$, blue) are near one-hot; soft labels ($T = 5$, red) reveal inter-class structure that guides the student.

(3) Inductive bias. The Hybrid's CNN head with kernel size 9 provides a strong structural prior for DNA motifs (TATA box $\sim$10 bp, CAAT box $\sim$8 bp). This constrains the student to learn biologically relevant features, making it more robust to noisy teacher signals.

6.2 Data Augmentation as Teacher Regularization

With augmentation, teacher accuracy drops from 96.6% to 91.8%, and KD becomes beneficial even for the 6-Layer student ($+0.3\%$). Augmentation prevents teacher overfitting, producing softer, more informative distributions. This aligns with Tang et al. Tang et al. [2020], who showed that teacher calibration is a key determinant of KD success. We recommend *always augmenting before distillation* on datasets with $<$20K training samples.

6.3 Cross-Species Generalization

Both distilled students outperform the teacher on mouse promoters (82.61% and 82.24% vs. 79.66%) despite lower human accuracy, suggesting that distillation acts as a *regularizer against species-specific overfitting*. The teacher memorizes human-specific promoter patterns, while the students' constrained capacity forces them to learn more generalizable motifs shared across mammals.

That the pure transformer—with no pretrained weights—nearly matches the DNABERT-2-initialized Hybrid on mouse data is particularly noteworthy: it implies the generalization benefit comes from the distillation process itself, not from the teacher's pretrained representations being inherited through weight initialization.

The near-random performance on plant promoters for all models confirms that mammalian promoter features do not transfer across kingdoms, consistent with the fundamental differences in plant vs. animal transcriptional regulation.

Table 6: 5-class multiclass results ($\alpha = 0.5$, $T = 3.0$). KD Δ = accuracy change from KD vs. baseline.

Setting	Model	Acc.	F1	Params	KD Δ
No Aug	Teacher	96.60	.966	117.1M	—
	6L Baseline	75.10	.741	46.7M	—
	6L + KD	73.70	.724	46.7M	-1.4
	Hybrid Base	83.00	.829	19.2M	—
	Hybrid + KD	86.80	.868	19.2M	$+3.8$
With Aug	Teacher	91.80	.918	117.1M	—
	6L Baseline	90.80	.908	46.7M	—
	6L + KD	91.10	.911	46.7M	$+0.3$
	3L Baseline	89.70	.897	25.4M	—
	3L + KD	89.30	.893	25.4M	-0.4
	Hybrid Base	87.30	.873	19.2M	—
	Hybrid + KD	**88.80**	.887	19.2M	$+1.5$

Table 7: Cross-species generalization. Drop = Human Acc − Cross-species Acc (lower is better).

Model	Human	Mouse	Plant	Drop	
	Acc	Acc	Acc	Mouse	Plant
Teacher (DNABERT-2)	96.71	79.66	48.22	17.04	48.48
Hybrid (init.)	93.31	**82.61**	47.48	**10.70**	45.83
4L-Pure-Trans. (KD)	93.01	82.24	46.20	10.76	46.80

6.4 Pure Architectures: KD Without Pretraining

The consistent $\sim$1% KD gain across all three pure architectures (Table 4) demonstrates that KD is effective even without pretrained initialization, though the absolute improvement is smaller than the $\sim$3.8% gains seen for pretrained Hybrids on multiclass tasks. The 4L-Pure-Transformer's 93.01% accuracy is only 0.30 points below the DNABERT-2-initialized Hybrid (93.31%), suggesting that for binary promoter classification, the teacher's soft-label guidance can largely compensate for the lack of pretrained weights.

The 4L-Pure-CNN's poor accuracy (84.83%) confirms that self-attention is essential for capturing long-range dependencies in promoter sequences, while the Hybrid-Pure-3T1C (92.85%) benefits from both local (CNN) and global (transformer) pattern detection.

6.5 Practical Deployment Considerations

The Hybrid student's 2.7$\times$ CPU speedup and 105 MB model size make it deployable on standard laptops—processing 1,000 sequences in $\sim$73 seconds on CPU vs. $\sim$197 seconds for the teacher. For laboratories without GPU access, this represents a meaningful practical improvement.

7 Conclusion

We presented a systematic study of knowledge distillation for compressing DNABERT-2 for DNA promoter classification. Our key findings are:

1. A **Hybrid CNN-Transformer** student (26.3 M params, 4.5$\times$ compression) achieves 93.31% accuracy—96.5% of the teacher—with 2.7$\times$ CPU speedup.
2. A **4L-Pure-Transformer** trained from scratch with KD achieves 93.01%, demonstrating that KD effectively transfers knowledge without pretrained initialization.
3. Both distilled students **outperform the teacher on mouse promoters** ($>$82% vs. 79.66%), showing that KD improves cross-species generalization.

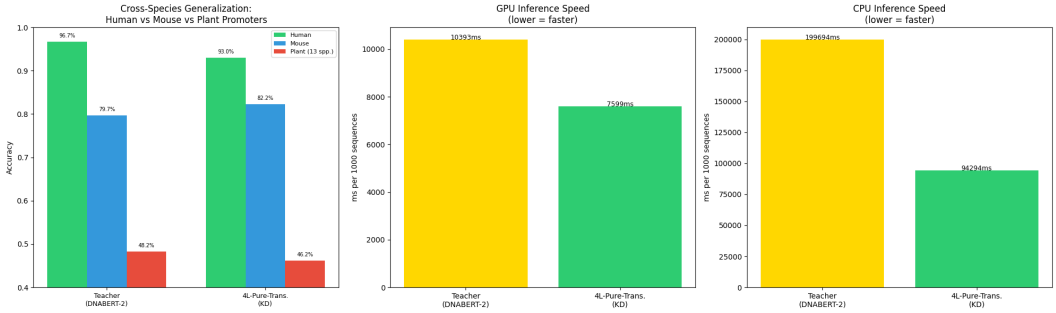

Figure 7: 4L-Pure-Transformer (KD) evaluation. **Left:** Cross-species accuracy (teacher vs. student). **Center:** GPU speed. **Right:** CPU speed.

Table 8: Inference speed (ms per 1,000 sequences, batch size 16).

Model	**GPU ms**	**CPU ms**	**GPU ↑**	**CPU ↑**
Teacher (DNABERT-2)	10,059	197,053	1.00×	1.00×
Hybrid (init.)	5,890	72,911	1.71×	**2.70**×
4L-Pure-Trans.	7,599	94,295	1.37×	2.12×

4. **Lower** α ($\alpha = 0.3$) and **moderate temperature** ($T = 3.0$) are optimal for the Hybrid architecture.
5. **Data augmentation is essential** for KD on small multiclass datasets, regularizing overconfident teachers to produce informative soft labels.

Future work includes adaptive per-sample temperatures, motif-aware intermediate distillation, and combining KD with INT8 quantization for further compression.

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